Q2 PART (A)

Statement: $\exists x(Course(x) \wedge \forall y(Student(y) \rightarrow Enrolled\_In(y, x)))$	$\forall x(\sim Course(x) \vee \exists y(Student(y) \wedge \sim Enrolled\_In(y, x)))$
Statement: $\forall x(Student(x) \rightarrow \exists y(Course(y) \wedge Enrolled\_In(x, y)))$	$\exists x(Student(x) \wedge \forall y(\sim Course(y) \vee \sim Enrolled\_In(x, y)))$
Statement: $\forall x((Teacher(x) \wedge Popular(x)) \rightarrow \exists y(Course(y) \wedge Popular(y) \wedge Teaches(x, y)))$	$\exists x((Teacher(x) \wedge Popular(x)) \wedge \forall y(\sim Course(y) \vee \sim Popular(y) \vee \sim Teaches(x, y)))$
Statement: $\exists x(Student(x) \wedge \forall y(Course(y) \rightarrow (Enrolled\_In(x, y) \wedge Smart(x))))$	$\forall x(\sim Student(x) \vee \exists y(Course(y) \wedge (\sim Enrolled\_In(x, y) \vee \sim Smart(x))))$

Q2 PART (B)

$\forall x(Course(x) \rightarrow \exists y\exists z(Student(y) \wedge Teacher(z) \wedge Enrolled\_In(y, x) \wedge Teaches(z, x)))$	$\exists x(Course(x) \wedge \forall y\forall z(\sim Student(y) \vee \sim Teacher(z) \vee \sim Enrolled\_In(y, x) \vee \sim Teaches(z, x)))$
$\exists x(Teacher(x) \wedge \forall y(Student(y) \rightarrow \forall z((Course(z) \wedge Teaches(x, z) \wedge Enrolled\_In(y, z)) \rightarrow Smart(y))))$	$\forall x(\sim Teacher(x) \vee \exists y(Student(y) \wedge \exists z(Course(z) \wedge Teaches(x, z) \wedge Enrolled\_In(y, z) \wedge \sim Smart(y))))$
$\forall x(Student(x) \rightarrow (Smart(x) \rightarrow \exists y(Teacher(y) \wedge Experienced(y) \wedge \exists z(Course(z) \wedge Teaches(y, z) \wedge Enrolled\_In(x, z)))))$	$\exists x(Student(x) \wedge Smart(x) \wedge \forall y(\sim Teacher(y) \vee \sim Experienced(y) \vee \forall z(\sim Course(z) \vee \sim Teaches(y, z) \vee \sim Enrolled\_In(x, z))))$
$\exists x(Course(x) \wedge \forall y((Teacher(y) \wedge Teaches(y, x)) \rightarrow Popular(y)) \wedge \forall z((Student(z) \wedge Enrolled\_In(z, x)) \rightarrow \forall w((Student(w) \wedge \sim Enrolled\_In(w, x)) \rightarrow Better\_Than(z, w))))$	$\forall x(\sim Course(x) \vee \exists y(Teacher(y) \wedge Teaches(y, x) \wedge \sim Popular(y)) \vee \exists z(Student(z) \wedge Enrolled\_In(z, x) \wedge \exists w(Student(w) \wedge \sim Enrolled\_In(w, x) \wedge \sim Better\_Than(z, w))))$

Q:3

m is odd so by def of odd number if k is any integer then $m = 2k + 1$

n is even so by def of even number if k is any integer then $n = 2k$

substitute the values in equation

$4m+3n = 4 (2k+1) + 3 (2k)$

$= 8K + 4 + 6k = 14K + 4 = 2 (7k + 2)$

Here $(7k + 2)$ is also an integer as multiplication and sum of integers is integer

Now as

$4m + 3n = 2(7k + 2)$

And $7k + 2$ can be written as Q

$Q = 7k + 2$

$4m + 3n = 2Q$

Here any integer multiplied with 2 is even so $4m + 3n$ is even